BioNexus Molecular Diagnostic Report

Multi-Omics Breast Cancer Subtype Classification (PAM50)

Case ID:	CASE-BRCA-4412	**Patient Ref:**	PATIENT-DEMO-002
Specimen:	primary_tumor	**Created:**	2026-10-02T19:15:00
Modalities:		**Model Version:**	1.0.0

Predicted PAM50 Subtype	Calibrated Confidence	Confidence Tier
ABSTAIN (Uncertain)	**54.8%**	**LOW_CONFIDENCE**

Subtype Probability Distribution

LumB	LumA	Her2	Normal	Basal
54.8%	38.2%	4.5%	1.5%	1.0%

Top Attributed Biomarkers (Integrated Gradients)

Cheapest-Next-Test Recommendations

Recommended Assay	Exp. Gain	Cost	Turnaround	Rationale
miRNA Sequencing	+8.2%	$350	1-2 days	Adding miRNA profiling provides highest subtype differentiation power to resolve LumA vs LumB boundary.
DNA Methylation	+4.5%	$450	5-7 days	Methylation status at ESR1/PGR promoter loci clarifies epigenetic silencing.

Input Quality Control Summary

Modality	Status	Mapped Features	Coverage %	Raw Scale Warning
RNA	WARNING	3950 / 5000	79.0%	NO

Validation Benchmark Summary (Stratified 5-Fold Nested CV)

Model benchmarked on locked TCGA-BRCA cohort (n=646). Overall fused performance: **86.99% ± 3.01% Accuracy**, **84.58% ± 2.96% Macro-F1**. Ablation baselines: RNA-only 81.04% F1, miRNA-only 72.42% F1, Methylation-only 72.80% F1. External calibration ECE: 0.048 across 15 bins.

Fingerprint: Git [4c267225b360] | Input Hash [sha256-f9e8d7c6b] | System: BioNexus GraphGPS v1.0.0

DISCLAIMER: Research use only, not a diagnostic device. BioNexus outputs are intended solely for investigative cancer research and hypothesis generation. Treatment decisions should not be based on these results without clinical pathology validation.